

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=0 success=False nfev=80 njev=6

message=The maximum number of function evaluations is exceeded.

reported optimality=1.2624563623830909e-05

initial={'cost': 861.254740620267, 'gradient_inf_norm': 0.0426175778391098, 'gradient_l2_norm': 0.06235242176017062, 'jacobian': {'condition_number_nonzero_subspace': 429.790154554383, 'numerical_rank': 14, 'numerical_tolerance': 2.1551810338576138e-10, 'singular_values': [226.77739488311005, 214.30129585984474, 175.24770669792971, 48.58633918565074, 44.66661191305999, 25.860229205276834, 13.131141924328379, 12.18761005143159, 11.742433227647608, 10.944721334478631, 9.769584111718158, 3.823490926142325, 3.7782333506665684, 0.5276467887409809]}, 'residual_l2_norm': 41.50312616226076}

final={'cost': 861.2547387045346, 'gradient_inf_norm': 1.2624563623830909e-05, 'gradient_l2_norm': 1.3004253257283628e-05, 'jacobian': {'condition_number_nonzero_subspace': 430.4615570220279, 'numerical_rank': 14, 'numerical_tolerance': 2.1551881121274534e-10, 'singular_values': [226.7781396890309, 214.3026209499731, 175.24669554926157, 48.586489861807316, 44.66666579363386, 25.860377259944425, 13.130820795909754, 12.18738952056772, 11.742540723844222, 10.94485492749247, 9.769788009515693, 3.8235620466500984, 3.77818764529506, 0.5268255341032139]}, 'residual_l2_norm': 41.50312611610202}

coordinate_step_l2_norm=0.0008234398350711468

cost_reduction=1.915732354973443e-06

gradient_tolerance_satisfied=False

finite_difference_max_relative_error=4.636641855461156e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 187.6432727786216, 'difference_l2_norm': 8.700346524610719e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 187.64327277856574, 'relative_error': 4.636641855461156e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.